Lecture 16: Summing Infinite Series I

Given a sequence $\{a_n\}$, the **series** associated with this sequence is the sum of the terms a_n.

$$\underbrace{a_1 + a_2 + \cdots + a_n} + \underbrace{a_{n+1} + \cdots} = \sum_{n=1}^{\infty} a_n \text{ : series.}$$

$$S_N = a_1 + a_2 + \cdots + a_N$$

$S_1 = a_1$

$S_2 = a_1 + a_2$

$S_3 = a_1 + a_2 + a_3$

is the Nth **partial sum** of a series.

$\boxed{\textbf{Sum of a Series}}$: is the limit of S_n.

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = s$$

We say the series **converges** to s if s is a finite number, otherwise, the series **diverges**.

ex. Suppose $\{a_n\} = 1$, determine the convergence of the series $\sum_{n=1}^{\infty} a_n$: dv. $S_n = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n$

$$\text{Sum} = \lim_{n\to\infty} S_n = \lim_{n\to\infty} n = \infty$$

ex. Given the sequence conv. to 0.

$$\{a_n\} = \left\{\frac{3}{10}, \frac{3}{100}, \frac{3}{1000}, \cdots\right\} = \left\{\frac{3}{10^n}\right\}_{n=1}^{\infty}, \qquad r = \frac{1}{10}.$$

what is the sum of the series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(\frac{3}{10^n}\right) = ?$$

Solution:

$S_1 = 3/10 = 0.3$

$S_2 = 3/10 + 3/100 = 0.3 + 0.03 = 0.33$

$S_3 = 3/10 + 3/100 + 3/1000 = 0.3 + 0.03 + 0.003 = 0.333$

$\vdots$

$S_N = 3/10 + 3/100 + \cdots + 3/10^N = 0.\underbrace{33\cdots3}_{N \text{ digits}}.$

$\lim_{N\to\infty} S_N = 0.33\cdots = 0.\overline{3} = \frac{1}{3}$

$$\sum_{n=1}^{\infty} \left(\frac{3}{10^n}\right) = \lim_{N\to\infty} S_N = \frac{1}{3}$$

A **Geometric series** is the summation of the terms of a geometric sequence $\{a, ar, ar^2, ar^3, \cdots\}$

first term

$$\sum_{n=0}^{\infty} ar^n = \boxed{a} + ar + ar^2 + ar^3 + \cdots = \frac{a}{1-r} \text{ if } |r|<1$$

$n=0$

$$= \boxed{\frac{\text{first term}}{1-r}}$$

.

Sum of Geometric Series

A geometric series $\sum_{n=1}^{\infty} ar^n = \frac{ar}{1-r}$, if $|r| < 1$ and diverges if $|r| \geq 1$.

pf:

$$S_N = ar + ar^2 + \cdots + ar^N$$

$$-) \quad rS_N = ar^2 + \cdots + ar^{N+1}$$

$$S_N - rS_N = ar - ar^{N+1}$$

$$S_N(1-r) = ar(1-r^N)$$

$r \neq 1$:

$$S_N = \frac{ar(1-r^N)}{1-r}$$

$$\text{sum} = \lim_{N\to\infty} S_N = \lim_{N\to\infty} \frac{ar(1-r^N)}{1-r} = \frac{ar}{1-r} \text{ if } |r|<1$$

$r^N \to 0$ if $-1<r<1$

Notice the numerator of this formula is always the **first term** of the series.

(ex. If $|r| < 1$, then, $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$)

$n=3: \frac{5}{2^3} = \frac{5}{8}$

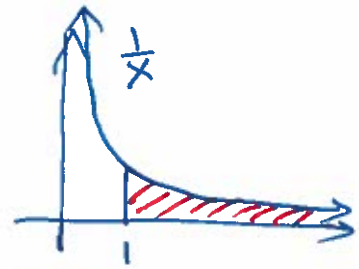
ex. Determine if the series $\sum_{n=3}^{\infty} \frac{5}{2^n}$ converges. If converges, find the sum (Geo., $\frac{5}{4}$)

$a_n = 5/2^n \to 0$

① Geo. series with $r = \frac{1}{2}$ $\longrightarrow$ conv.

② $\text{Sum} = \frac{\text{first term}}{1-r} = \frac{5/8}{1-1/2} = \frac{5/8}{1/2} = 5/4$

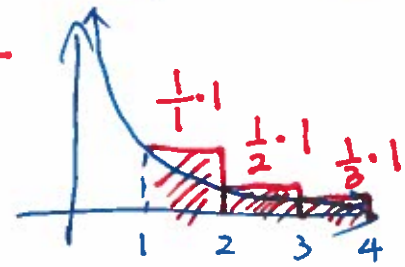
$\int_1^\infty \frac{1}{x}\,dx$ div.



Harmonic Series

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n} : \text{div}}$$

$a_n = \frac{1}{n} \to 0$



$$S_1 = 1$$
$$S_2 = 1 + \frac{1}{2}$$
$$S_3 = 1 + \frac{1}{2} + \frac{1}{3}$$
$$S_{2^2} = S_4 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + 2\cdot\frac{1}{2}$$
$$\vdots$$
$$S_{2^3} = S_8 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$
$$\vdots$$
$$> 1 + 3\cdot\frac{1}{2}$$
$$S_{2^N} > 1 + N\cdot\frac{1}{2}$$
$$\sum_{n=1}^{\infty} \frac{1}{n} = \lim_{n\to\infty} S_{2^N} > \lim_{N\to\infty} 1 + N\cdot\frac{1}{2} = \infty$$

Common Mistake – Just because the terms a_n decrease to 0 does NOT mean the series $\sum a_n$ converges. For instance, the harmonic series diverges even though its terms $\frac{1}{n}$ decrease to zero.

$\lim_{n\to\infty} S_n = S < \infty$ exists.

$\Uparrow$

Theorem: If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

$\lim_{n\to\infty}(S_n - S_{n-1} = a_n)$

$0 = S - S = \lim_{n\to\infty} a_n$

*** **nth-Term Test (for Divergence):**

If $\lim_{n\to\infty} a_n \neq 0$, then the series diverges.

Warning: This test does NOT work in REVERSE. If $\lim_{n\to\infty} a_n = 0$, we can draw NO conclusions about its convergence or divergence.

NYTI: Determine if the series converges or diverges. If converges, find the sum.

- $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$ div. TFD. $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n^2}{5n^2+4} = \frac{1}{5} \neq 0$
- $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} \cdots$ conv. : Geo. series with $\left|r = -\frac{2}{3}\right| < 1$

$$\text{Sum} = \frac{5}{1-(-2/3)} = \frac{5}{5/3} = 3$$

$(\ln n)^q \ll n^p \ll a^n \ll n! \ll n^n$

- $\sum_{n=1}^{\infty} \frac{e^n}{n^2}$ div. by TFD. $\lim_{n\to\infty} \frac{e^n}{n^2} = \infty \neq 0$
- $\sum_{n=1}^{\infty} (-1)^n$ div. by TFD. $\lim_{n\to\infty} (-1)^n = \text{DNE} \neq 0$

Theorem: If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

nth-Term Test (for Divergence):

If $\lim_{n\to\infty} a_n \neq 0$, then the series diverges.

Warning: This test does NOT work in REVERSE. If $\lim_{n\to\infty} a_n = 0$, we can draw NO conclusions about its convergence or divergence.

NYTI: Determine if the series converges or diverges. If converges, find the sum.

- $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$ div by TFD. $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n^2}{5n^2+4} = \frac{1}{5} \neq 0$
- $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} \cdots$: Geo series with $\left|r = -\frac{2}{3}\right| < 1$

  Conv. to 3.

  $\text{sum} = \frac{\text{first term}}{1-r} = \frac{5}{1-(-\frac{2}{3})} = \frac{5}{5/3} = 3$

$(\ln n)^q \ll n^p \ll a^n \ll n! \ll n^n$

- $\sum_{n=1}^{\infty} \frac{e^n}{n^2}$ div by TFD $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{e^n}{n^2} = \infty \neq 0$
- $\sum_{n=1}^{\infty} (-1)^n$ div. by TFD $\lim_{n\to\infty} (-1)^n = \text{DNE} \neq 0$

- $\sum_{n=1}^{\infty}(1/2)^{n-1}$ conv. to 2. (Geometric, 2) $\left|r=\frac{1}{2}\right|<1$, sum $=\frac{\text{first term}}{1-r}=\frac{1}{1-\frac{1}{2}}=2$

- $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n}$ div by TFD $\lim_{n\to\infty} a_n=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e\neq 0$

- $\sum_{n=3}^{\infty}(-5/4)^{n}$ div. (Geo., Diverges) $\left|r=-\frac{5}{4}\right|>1$

- $\sum_{n=1}^{\infty}2^{2n}3^{1-n}=\sum_{n=1}^{\infty}(2^2)^n\cdot\frac{3}{3^n}=\sum_{n=1}^{\infty}3\left(\frac{2^2}{3}\right)^n=\sum_{n=1}^{\infty}3\cdot\left(\frac{4}{3}\right)^n$ div.

  $\left|r=\frac{4}{3}\right|>1$

- $\sum_{n=1}^{\infty}\frac{e^n}{3^{n-1}}$ (Geo., $\frac{3e}{3-e}$) $=\sum_{n=1}^{\infty}\frac{e^n}{3^{-1}\cdot 3^n}=\sum_{n=1}^{\infty}3\cdot\left(\frac{e}{3}\right)^n$ conv. to $\frac{3e}{3-e}$

  $\left|r=\frac{e}{3}\right|<1$ sum $=\frac{e}{1-\frac{e}{3}}=\frac{3e}{3-e}$

- $\sum_{n=1}^{\infty}\sqrt[n]{2}$ div. (TFD) $\lim_{n\to\infty}a_n=\lim_{n\to\infty}\sqrt[n]{2}=\lim_{n\to\infty}2^{1/n}=2^0=1\neq 0$

- $\sum_{n=1}^{\infty}(-1)^n\frac{n^2}{11n^2+1}$ div by TFD.

  $\lim_{n\to\infty}|a_n|=\lim_{n\to\infty}\frac{n^2}{11n^2+1}=\frac{1}{11}$, so $\lim_{n\to\infty}a_n$ DNE $\neq 0$

- If you delete a <u>finite</u> number of terms from a divergent series, will the new series still diverge? Yes. explain your reasoning.

- If you add a finite number of terms to a convergent series, will the new series still converge? Yes. Explain your reasoning.

* NOTE: Whether a series converges does not depend on a finite number of terms added to or removed from the series. However, the <u>value</u> of a convergent series does change if nonzero terms are added or deleted.

- If $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ converges. (T/F)

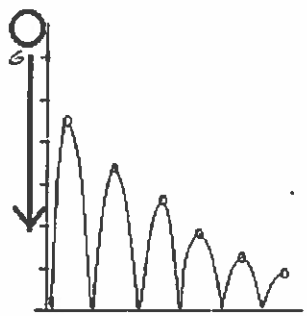
- If $|r| < 1$, then $\sum_{n=1}^{\infty} ar^n = \frac{a}{1-r}$. (T/F) Sum $= \frac{ar}{1-r}$

- The series $\sum_{n=1}^{\infty} \frac{n}{1000(n+1)}$ diverges. (T/F) $\lim_{n\to\infty} a_n = \frac{1}{1000} \neq 0$

conv. if $\lim_{n\to\infty} a_n$ exists. conv. if $\lim_{n\to\infty} S_n$ exists

	$\{a_n\} =$ sequence	$\sum a_n =$ series
$\lim a_n = 0$	conv. to 0	? $\sum \frac{1}{n}$ div. $\sum (\frac{1}{2})^n$ conv.
$\lim a_n = c \overset{<\infty}{\neq} 0$	conv. to c.	div. by TFD.
$\lim S_n = 0$		conv. to 0
$\lim S_n = c \overset{<\infty}{\neq} 0$		conv. to c.

- A ball is droopped from a height of 6 feet and begins bouncing. The height of each bounce is 3/4th the height of the previous bounce. Find the **total vertical distance** traveled by the ball.



$$D = D_1 + D_2 + D_3 + \cdots$$

$$= 6 + 2(6(3/4)) + 2(6(3/4)(3/4)) + \cdots$$

$$= 6 + 12(3/4) + 12(3/4)^2 + 12(3/4)^3 + \cdots$$

$$= 6 + 12(\sum_{n=1}^{\infty}(3/4)^n) = 42$$